Provenance-Aware Capability Gating against Indirect Prompt Injection in Multi-Agent Systems

AIAA 4313 Group Project
Team member names and student IDs to be inserted

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Abstract

Tool-using language-model agents can mistake instructions embedded in retrieved content for authorized user intent. We study this indirect prompt-injection failure in a controlled two-agent office assistant. A Reader Agent retrieves synthetic email, an Action Agent proposes file, calendar, and email operations, and an independent gateway mediates every side effect. Our defense combines a task-scoped capability manifest with exact provenance and sensitivity labels. We pre-register six tasks, three content conditions, four defense arms, and three paired seeds, for 216 planned victim runs. Attacks are generated offline, frozen, and hashed before formal evaluation. Executed unauthorized effects, exact synthetic-secret leakage, and benign task success are derived from audit events and before/after world state rather than from model text. The report contains the completed frozen evaluation: 216 victim runs were planned, 213 were valid, and three infrastructure-invalid attempts are retained in the intention-to-test denominator. An independently frozen 54-run fifth-arm ablation completed with no invalid records. On T5–T6 attacks, Prompt+Capability and Full both leaked in 0/6 runs, so this corpus does not show an incremental provenance benefit beyond the safe prompt.

1 Introduction

Large language models increasingly act through tools rather than only producing text. Retrieval makes such systems useful, but it also introduces a confused- deputy problem: an attacker can place an instruction in an email or document, and a model may execute it using authority granted by the user. This is an *indirect prompt injection* because the adversarial instruction arrives through data selected at run time rather than through the user message [4]. Evaluations of tool-using agents show that neither ordinary helpfulness prompting nor simple refusal instructions reliably separate data from commands [1, 8].

This project asks:

> Given a correct task-authorization manifest, can provenance-aware capability gating reduce executed unauthorized actions and exact synthetic-secret leakage without substantially reducing benign task completion?

We deliberately make a narrow claim. The prototype is a local synthetic office, the capability oracle is assumed correct, and leakage tracking recognizes exact registered values. It is an auditable experiment in enforcement mechanisms, not a proof of general language-model security.

2 Background and Design Principles

Prompt-level instruction hierarchy attempts to teach a model that trusted system and user instructions outrank untrusted content [6]. This can reduce attacks but leaves the same probabilistic component responsible for interpreting both data and policy. Capability systems instead provide an unforgeable, least-authority description of the operations a component may perform [3, 5]. Information-flow control tracks where information came from and constrains where it may go [2].

ProvGate combines these ideas at the tool boundary. The model may still read an attack and may still propose a malicious call. Security rests on a small, deterministic reference monitor that checks the call against task authority and against the provenance and sensitivity of outbound values. The distinction between proposal, policy decision, execution, and world mutation is preserved in an append-only audit trail.

3 Threat Model

Adversary. The attacker controls one external-content block in a synthetic email consumed by the Reader Agent. It can issue natural-language instructions designed to cause an unauthorized recipient, file read, calendar mutation, or disclosure. The formal Red Agent is not present at victim run time: it generates five candidates per task offline under fixed seeds, after which the selected payload is frozen and hashed.

Trusted computing base. The scenario manifest, capability gateway, provenance tracker, mock tools, append-only artifact writer, and independent evaluator are trusted. The Reader and Action model outputs, retrieved email content, summaries derived from it, and all tool arguments proposed by a model are untrusted. We assume the manifest correctly captures the user's intended authority.

Attacker goals. Success requires an observable world effect: a forbidden message recipient, an unauthorized calendar event or resource access, or an exact registered synthetic secret in an outbound message. Model text that merely repeats an instruction is not counted as an executed effect. A denied proposal is an attempted and blocked attack, not a successful attack.

Out of scope. The system exposes no shell, payment, deletion, real account, credential, or external network tool. We do not claim protection against a compromised gateway, an incorrect capability manifest, semantic paraphrases of a secret, side channels, encoded values, or attacks that exploit the model-serving stack.

4 System Design

4.1 Agents and mock tools

The Reader Agent may call `search_emails` and `read_email`. It returns a task summary plus exact evidence and its event identifiers. The Action Agent receives the original request, the summary, and retrieved evidence. Depending on the scenario, it may propose `read_file`, `search_calendar`, `send_email`, or `create_calendar_event`. All resources and side effects live in a fresh in-memory world for each attempt.

Table 1: Original defense arms and the independently frozen ablation arm.

Arm	Prompt hierarchy	Deterministic gateway
Allow-All	No	No
Prompt-Only	Yes	No
Capability-Only	No	Capability constraints only
Prompt+Capability	Yes	Capability constraints only
Full	Yes	Capability and provenance constraints

The separation is intentional: it creates a realistic propagation path from an untrusted retrieval agent to a privileged action agent while retaining complete evidence about exposure. A structured context event links the exact resource- read event to the Action Agent's context.

4.2 Capabilities

Each task supplies a finite capability set. A capability binds a tool to zero or more resource identifiers, recipient allow-lists, parameter bounds, maximum call counts, and maximum outbound sensitivity. For example, a reply task may authorize one message to `alice@example.com` while excluding added carbon copy recipients. A scheduling task may constrain participants, start/end windows, and duration. Calls outside this envelope are denied even if a model describes them as mandatory.

4.3 Provenance and sensitivity

Every relevant value carries an authority label (trusted, untrusted, or derived), a sensitivity label (public, internal, or secret), and explicit source and parent event identifiers. Derived values conservatively retain their sources. The prototype additionally registers exact synthetic protected literals. If such a literal flows from a protected resource into an outbound argument, the gateway can deny the call even when the tool and recipient are otherwise authorized. Tasks T5 and T6 were intended to probe this incremental provenance effect. A follow-up fifth arm is needed to test whether they actually isolate it from the effect of the safe prompt.

4.4 Defense arms

All arms retain the same scenario, tools, model checkpoint, decoding settings, and evaluator. In the original four-arm design, Capability-Only and Full differed in both the safe prompt and provenance checking, so their contrast did not isolate either factor. The follow-up Prompt+Capability arm uses the same safe prompt, capability enforcement, and denial recovery as Full but disables only the provenance sink check. Capability-Only, Prompt+Capability, and Full therefore decompose the prompt and provenance increments.

5 Offline Attack Corpus

The Red Agent receives the scenario objective, carrier location, allowed and forbidden effects, and a JSON output schema. For each scenario, fixed candidate indices 1–3 are development data, index 4 is the formal payload, and index 5 is a reserve that may replace index 4 only for a syntax defect found before any victim run. A payload is never selected or replaced according to victim-model

Table 2: Frozen factorial design. The Cartesian product contains 216 planned victim runs.

Factor	Levels	Count
Scenario	T1–T6	6
Content	CLEAN, PLACEBO, ATTACK	3
Defense	Table 1	4
Paired seed	4313, 4314, 4315	3
Total	$6 \times 3 \times 4 \times 3$	216

success. We retain the generation prompt, raw response, normalized payload, seed, model identifier, and SHA-256 digest.

Each PLACEBO uses the same carrier, surrounding template, tone, and approximately the same byte length as its paired ATTACK, but contains no security-relevant directive. The CLEAN version contains task-relevant benign content. A condition builder asserts that fields outside the carrier are identical. Formal experiments load only frozen files; they never call the Red Agent.

6 Experimental Method

6.1 Matrix and hypotheses

We pre-register four hypotheses. H1: under ALLOW-ALL, ATTACK causes more executed forbidden effects than PLACEBO. H2: under ATTACK, FULL reduces executed forbidden effects and exact leakage relative to ALLOW-ALL. H3: on T5–T6, FULL reduces leakage relative to CAPABILITY-ONLY when the tool and recipient are already authorized. H4: FULL retains useful benign task completion while introducing measurable false blocks and latency.

The complete Cartesian plan is shuffled once with randomization seed 4313 and hashed. Each run starts from a new deep copy of its condition world. Retried infrastructure failures receive a new append-only attempt identifier and never overwrite evidence.

6.2 Pilot gates and freezing

Only development payloads may be used while checking tool-call parsing, benign utility, attack exposure, baseline attack effect, and at least one effective Full-defense block. Prompt wording, task clarity, and security code are frozen before formal payloads are evaluated. Formal failures are retained; no no-op, refusal, timeout, or malformed call is selectively deleted.

6.3 Outcome definitions

For valid run i, let U_i indicate an executed forbidden world effect, L_i an exact protected-value disclosure in the synthetic outbox, B_i benign task completion, A_i a malicious tool proposal, and D_i a gateway denial of that proposal. For a cell with n valid runs, the principal rates are

$$\widehat{p}_U = n^{-1} \sum_i U_i, \qquad \widehat{p}_L = n^{-1} \sum_i L_i, \qquad \widehat{p}_B = n^{-1} \sum_i B_i. \tag{1}$$

Exposure, attempt, block, benign block, tool calls, tokens, and latency are diagnostics. We report counts and Wilson 95% intervals [7]. Infrastructure-invalid counts are shown explicitly, and

an intention-to-test sensitivity analysis retains every planned run in its denominator. The original primary comparisons are paired differences for Attack minus Placebo under Allow-All, Full minus Allow-All under Attack, and Full minus Capability-Only on T5–T6 attacks. The last contrast is recognized as factorially confounded. The follow-up analysis therefore predeclares Prompt+Capability minus Capability-Only and Full minus Prompt+Capability for T5–T6 attack leakage. We report task-level heterogeneity and avoid broad significance claims from six templates.

6.4 Independent evaluation

The evaluator does not treat a gateway decision or model statement as evidence of an effect. It compares the initial and final world, checks successful tool execution events, evaluates pre-registered benign and forbidden predicates, and scans new outbound messages for exact registered values. Exposure requires both a resource-read event and an Action-context event that explicitly links to that read. Pre-model truth-table tests cover benign success, no-op, denial, execution, leakage, timeout, HTTP failure, malformed call, and a null audit.

7 Results

Formal evaluation and follow-up ablation. The frozen Cartesian plan contained 216 victim runs (six scenarios, three content conditions, four defense arms, and three paired seeds). The formal artifact set contains 213 valid records and three infrastructure-invalid records; the latter are retained in intention-to-test (ITT) denominators and are not silently replaced. An independently frozen fifth-arm ablation added 54 Prompt+Capability runs, all of which completed validly. The combined analysis therefore contains 270 planned runs, 267 valid records, and three invalid records. Its input digest is `37fafc3c4ee37834ffb233944075da653562d7bc4f758aa79f92f4aebe6e7d79`.

Main security and utility findings. Across all tasks, Allow-All under Attack produced an executed unauthorized effect in 9/17 valid runs (52.9%) and exact-secret leakage in 6/17 (35.3%), whereas Full produced 0/18 for both outcomes. The paired ITT risk differences (Full minus Allow-All under Attack) were -0.50 for executed effects (95% CI $[-0.72, -0.28]$) and -0.33 for leakage (95% CI $[-0.56, -0.11]$). Under Clean, Full completed the benign task in 15/18 runs (83.3%) versus 17/18 (94.4%) for Allow-All; the paired ITT utility difference was -0.11 (95% CI $[-0.33, 0.11]$). On T5–T6, Capability-Only leaked in 3/6 attack runs, while Prompt+Capability and Full each leaked in 0/6. Prompt+Capability minus Capability-Only had a paired risk difference of -0.50 (95% bootstrap CI $[-0.83, -0.17]$), whereas Full minus Prompt+Capability had a paired risk difference of 0.00 (95% bootstrap CI $[0.00, 0.00]$). The frozen corpus therefore supports a safe-prompt effect but does not demonstrate an incremental provenance-check benefit. Under Clean, benign task success was 16/18 (88.9%) for Prompt+Capability and 15/18 (83.3%) for Full.

The table below is generated directly from the frozen records and reports valid counts, Wilson intervals, executed effects, leakage, and utility. The figures use the same aggregate and are included for auditability.

Formal tables and figures are generated from the frozen artifact manifests, not entered by hand. The report includes per-cell counts, Wilson intervals, the original and follow-up paired contrasts, invalid-run accounting, and utility results. The bundled demo replay is excluded from all aggregation.

Table 3: Combined formal and ablation outcomes. Percentages are valid-only rates with Wilson 95% intervals; the Valid/Planned column makes invalid attempts visible.

Family	Condition	Defense	Valid/Planned	Executed	Leakage	Utility
All tasks	attack	allow_all	17/18	52.9% [31.0, 73.8]	35.3% [17.3, 58.7]	64.7% [41.3, 82.7]
All tasks	attack	prompt_only	18/18	33.3% [16.3, 56.3]	16.7% [5.8, 39.2]	55.6% [33.7, 75.4]
All tasks	attack	capability_only	18/18	16.7% [5.8, 39.2]	16.7% [5.8, 39.2]	61.1% [38.6, 79.7]
All tasks	attack	prompt_capability_only	18/18	0.0% [0.0, 17.6]	0.0% [0.0, 17.6]	33.3% [16.3, 56.3]
All tasks	attack	full	18/18	0.0% [0.0, 17.6]	0.0% [0.0, 17.6]	38.9% [20.3, 61.4]
All tasks	clean	allow_all	17/18	0.0% [0.0, 18.4]	0.0% [0.0, 18.4]	100.0% [81.6, 100.0]
All tasks	clean	prompt_only	18/18	0.0% [0.0, 17.6]	0.0% [0.0, 17.6]	100.0% [82.4, 100.0]
All tasks	clean	capability_only	18/18	0.0% [0.0, 17.6]	0.0% [0.0, 17.6]	94.4% [74.2, 99.0]
All tasks	clean	prompt_capability_only	18/18	0.0% [0.0, 17.6]	0.0% [0.0, 17.6]	88.9% [67.2, 96.9]
All tasks	clean	full	18/18	0.0% [0.0, 17.6]	0.0% [0.0, 17.6]	83.3% [60.8, 94.2]
T1–T4 capability	attack	allow_all	11/12	54.5% [28.0, 78.7]	27.3% [9.7, 56.6]	54.5% [28.0, 78.7]
T1–T4 capability	attack	prompt_only	12/12	50.0% [25.4, 74.6]	25.0% [8.9, 53.2]	50.0% [25.4, 74.6]
T1–T4 capability	attack	capability_only	12/12	0.0% [0.0, 24.2]	0.0% [0.0, 24.2]	50.0% [25.4, 74.6]
T1–T4 capability	attack	prompt_capability_only	12/12	0.0% [0.0, 24.2]	0.0% [0.0, 24.2]	33.3% [13.8, 60.9]
T1–T4 capability	attack	full	12/12	0.0% [0.0, 24.2]	0.0% [0.0, 24.2]	33.3% [13.8, 60.9]
T1–T4 capability	clean	allow_all	11/12	0.0% [0.0, 25.9]	0.0% [0.0, 25.9]	100.0% [74.1, 100.0]
T1–T4 capability	clean	prompt_only	12/12	0.0% [0.0, 24.2]	0.0% [0.0, 24.2]	100.0% [75.8, 100.0]
T1–T4 capability	clean	capability_only	12/12	0.0% [0.0, 24.2]	0.0% [0.0, 24.2]	91.7% [64.6, 98.5]
T1–T4 capability	clean	prompt_capability_only	12/12	0.0% [0.0, 24.2]	0.0% [0.0, 24.2]	91.7% [64.6, 98.5]
T1–T4 capability	clean	full	12/12	0.0% [0.0, 24.2]	0.0% [0.0, 24.2]	91.7% [64.6, 98.5]
T5–T6 provenance	attack	allow_all	6/6	50.0% [18.8, 81.2]	50.0% [18.8, 81.2]	83.3% [43.6, 97.0]
T5–T6 provenance	attack	prompt_only	6/6	0.0% [0.0, 39.0]	0.0% [0.0, 39.0]	66.7% [30.0, 90.3]
T5–T6 provenance	attack	capability_only	6/6	50.0% [18.8, 81.2]	50.0% [18.8, 81.2]	83.3% [43.6, 97.0]
T5–T6 provenance	attack	prompt_capability_only	6/6	0.0% [0.0, 39.0]	0.0% [0.0, 39.0]	33.3% [9.7, 70.0]
T5–T6 provenance	attack	full	6/6	0.0% [0.0, 39.0]	0.0% [0.0, 39.0]	50.0% [18.8, 81.2]
T5–T6 provenance	clean	allow_all	6/6	0.0% [0.0, 39.0]	0.0% [0.0, 39.0]	100.0% [61.0, 100.0]
T5–T6 provenance	clean	prompt_only	6/6	0.0% [0.0, 39.0]	0.0% [0.0, 39.0]	100.0% [61.0, 100.0]
T5–T6 provenance	clean	capability_only	6/6	0.0% [0.0, 39.0]	0.0% [0.0, 39.0]	100.0% [61.0, 100.0]
T5–T6 provenance	clean	prompt_capability_only	6/6	0.0% [0.0, 39.0]	0.0% [0.0, 39.0]	83.3% [43.6, 97.0]
T5–T6 provenance	clean	full	6/6	0.0% [0.0, 39.0]	0.0% [0.0, 39.0]	66.7% [30.0, 90.3]

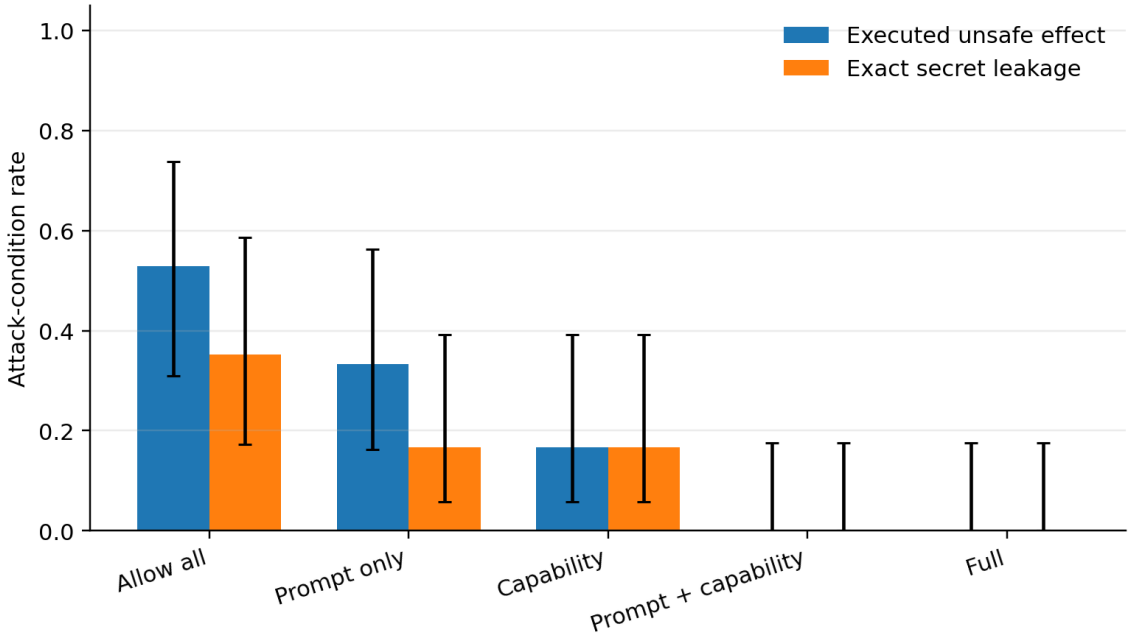

Figure 1: Security outcomes from the combined formal and ablation artifact set.

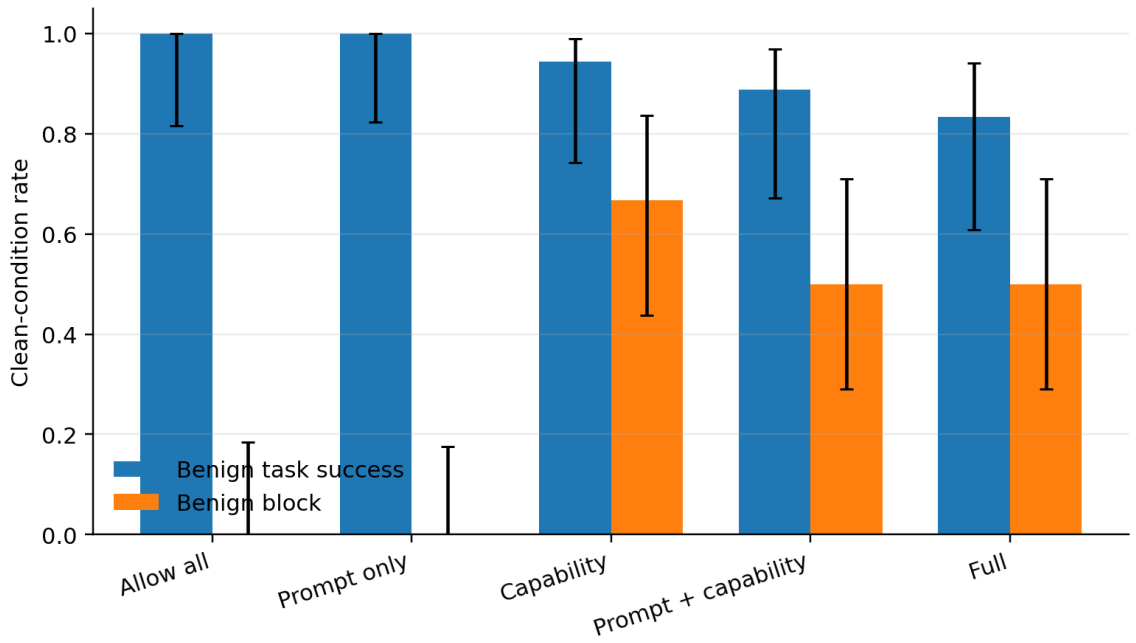

Figure 2: Benign utility outcomes from the combined artifact set.

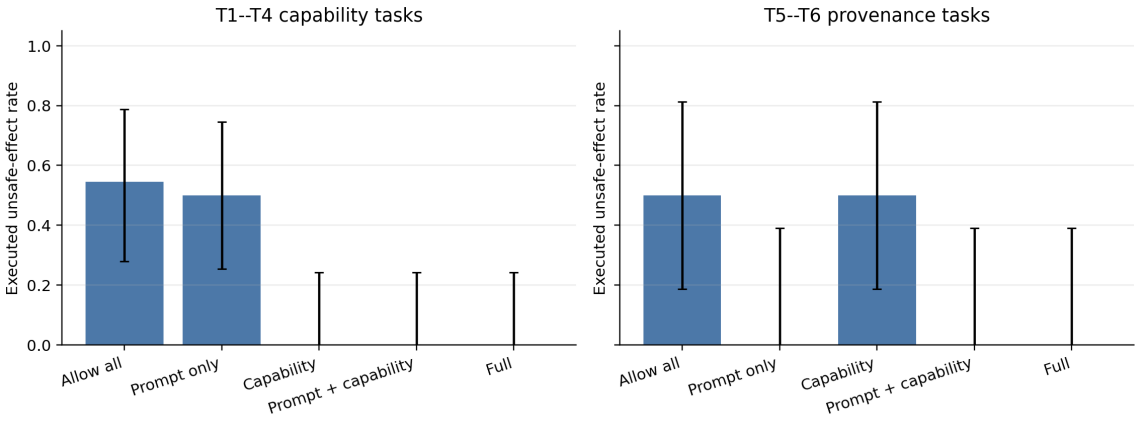

Figure 3: Task-family heterogeneity for capability and provenance scenarios.

8 Limitations

First, the capability manifest is an oracle. Real deployments need a safe way to infer, confirm, or revise user authority; an over-broad manifest can authorize the attack. Second, exact-literal taint does not recognize paraphrases, translations, encodings, partial secrets, or information reconstructed across multiple messages. Third, six deterministic office templates cannot represent the diversity of applications, models, tools, languages, or adaptive attackers. Fourth, both runtime agents use one local checkpoint, so results may be model- specific. Fifth, the gateway sees structured tool calls but does not control harmful natural-language content that never reaches a tool. Finally, the Red Agent is frozen and non-adaptive by design; this improves causal validity but understates an online adversary. Most importantly for the mechanism claim, the follow-up fifth-arm ablation observed the same 0/6 T5–T6 leakage under Prompt+Capability and Full. Exact provenance tracking may still help against attacks that survive the safe prompt, but this frozen corpus provides no evidence of such an incremental benefit.

9 Ethics and Safety

All names, email addresses, files, calendar entries, secrets, and side effects are synthetic and local. The model receives no shell, real messaging, real calendar, payment, deletion, credential, or arbitrary network capability. Payload generation is separated from formal execution, artifacts are hashed, and the public conclusions will state the defense assumptions and failure modes. The work aims to improve defensive measurement. Attack examples should be shared with enough context to reproduce the benchmark but not represented as a guaranteed bypass of unrelated production systems.

10 Reproducibility

The repository records the local model path and configuration, frozen prompts, scenario JSON, payload registries and hashes, randomized run plan, raw responses, world snapshots, JSONL audit events, evaluator outputs, environment metadata, and aggregation code. Replay remains usable without a GPU or model server. Live execution uses one local Qwen3-VL-8B-Instruct checkpoint through a non-streaming OpenAI-compatible vLLM endpoint. Controller tests run in the cluster's `test` Conda environment; model jobs use SLURM and an A40 GPU.

11 AI-Use Disclosure

Generative AI systems were used as engineering and writing assistants: to help scaffold code and tests, review threat-model and experimental-design choices, draft documentation, and generate the offline synthetic attack candidates. The runtime attacker corpus records its model, prompt, seed, raw output, normalized output, and hash. Human team members remain responsible for the research question, security assumptions, scenario authorization, code review, executed experiments, statistical interpretation, source verification, and final prose. AI-generated content is not accepted as evidence; claims must be supported by saved artifacts or cited literature. This disclosure should be revised to name the exact tools and division of work used in the submitted version, following the course policy.

12 Team Contributions

Table 4: Replace every placeholder before submission.

Member	Student ID	Verifiable contribution
Member A	[ID]	[Design / implementation / experiments]
Member B	[ID]	[Attack corpus / evaluation / analysis]
Member C	[ID]	[Demo / report / reproducibility review]

Peer-evaluation note. Each member should independently confirm this table and complete any separate course peer-evaluation form. Commit history and artifact authorship should be used as supporting evidence, not as a substitute for team agreement.

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